

ISLAMIC UNIVERSITY OF TECHNOLOGY



DESIGN PROJECT REPORT

CSE 4610 : Design Project

Deep Reinforcement Learning for Autonomous Vehicle Navigation at Unsignalized Intersections: Comparative Reproduction, Intent Encoding, and Sim-to-Sim Dynamics

Submitted by:

Farhan Tahsin Khan

220041229

Mahmudul Hasan

220041231

A.K.M Ishmam Tahmid

220041259

Supervised by:

Supervisor Name

Professor / Associate Professor, Department of Computer Science and Engineering

Department of Computer Science and Engineering

Islamic University of Technology (IUT)

October, 2026

Contents

1	Introduction	1
1.1	Background and Motivation	1
1.2	Complex Engineering Problem (CEP) Justification	1
1.3	Problem Statement and Research Objectives	2
1.4	Research Challenges and Main Contributions	2
1.5	Report Organization	2
2	Related Works	3
2.1	Introduction to Autonomous Intersection Management	3
2.2	The Five-Axis AIM Taxonomy	3
2.3	DRL for Flow Stabilization: Insights from Ring Networks	4
2.4	Unsignalized Intersection Navigation and the Multi-Task Dilemma	5
2.5	Identified Research Gaps	5
3	Problem Definition and Scope	6
3.1	Intersection Geometry and Spatial Configuration	6
3.2	Multi-Task Trajectory and Conflict Point Topology	6
3.3	Partially Observable Markov Decision Process (POMDP) Model	7
3.4	Defined Project Scope and Boundaries	7
4	Proposed Methodology	8
4.1	System Architecture and Control Pipeline	8
4.2	Observation State Space Formulations	8
4.3	Action Mapping and Proportional Controller	9
4.4	Comparative Evaluation of State-of-the-Art RL Algorithms	9
5	System Implementation	11
5.1	Simulation Infrastructure and Flow-SUMO Integration	11
5.2	Algorithmic Implementation Paradigms	11
5.3	Hardware Optimization: P-Core Affinity Pinning	12
6	Results and Discussion	13
6.1	Introduction and Master Comparative Results	13
6.2	Phase 2 Training Dynamics and Alpha Adaptation	13
6.3	Deterministic Evaluation and Behavioral Impact of Task Intent (μ_a)	14
6.4	Phase 1 PPO Baseline Experimental Evaluation	14
6.5	Critical Scientific Discussion: Demystifying the 99.52% Success Rate	15
7	Ethical, Social, and Environmental Considerations	16
7.1	Ethical Dimensions: Scientific Integrity and Fairness	16
7.2	Societal Impacts: Bottleneck Relief and Safety	16
7.3	Environmental Sustainability and Resource Efficiency	16
8	Challenges and Limitations	17
8.1	Physical Geometry and Inflow Saturation	17
8.2	Sim-to-Sim Disparities and Yielding Shields	17
8.3	Telemetry Sensitivity and Deployment Boundaries	17
9	Conclusion and Future Work	18
9.1	Summary of Completed Research	18
9.2	Architectural Recommendations for Future Work	18
9.3	Concluding Remarks	18
	References	19
	Appendices	20
A	Weekly Meeting Summary	20
B	Code and Implementation Artifacts	21
B.1	Quantitative Summary: GitHub Metrics	21
B.2	Feature Ownership and Technical Contribution	21
B.3	Verification: GitHub Screenshots	21

List of Figures

2.1	Hierarchical Taxonomy of Autonomous Intersection Management across Five Structural Dimensions.	4
2.2	Shockwave Attenuation and DRL Velocity Regularization in Benchmark Ring Networks.	4
4.1	Closed-Loop System Architecture and POMDP Control Pipeline linking TraCI/-SUMO to Actor-Critic Policies.	8
6.1	Phase 2 Discrete SAC Training Convergence and Dynamic Entropy Temperature Adaptation.	14
6.2	Deterministic Multi-Task Evaluation Metrics for Discrete SAC under Microscopic Simulation.	14
B.1	GitHub Repository Verification Telemetry Artifacts.	21

List of Tables

2.1	Comparative Synthesis of Autonomous Intersection Management Literature and Project Phases	5
3.1	Topological Conflict Point Characterization across Multi-Task Maneuvers . . .	6
4.1	Comparative Multi-Criteria Decision Analysis of Reinforcement Learning Algorithms	10
5.1	Hyperparameter Alignment Matrix between Xiao et al. (2024) and Our Phase 2 Reproduction	12
5.2	Computational Benchmark: Default OS Scheduling vs. P-Core Affinity Pinning	12
6.1	Master Comparative Results Matrix benchmarking Xiao et al. (2024), Phase 1 PPO, and Phase 2 Discrete SAC	13
A.1	Group Meeting and Milestone Contribution Table	20
B.1	Developer Contribution Metrics (GitHub Repository Telemetry)	21

List of Abbreviations

AEV	Autonomous Electric Vehicle
AIM	Autonomous Intersection Management
CAV	Connected and Automated Vehicle
CEP	Complex Engineering Problem
DQN	Deep Q-Network
DRL	Deep Reinforcement Learning
FCFS	First-Come-First-Served
IDM	Intelligent Driver Model
MDP	Markov Decision Process
MLP	Multi-Layer Perceptron
OpenMP	Open Multi-Processing
POMDP	Partially Observable Markov Decision Process
PPO	Proximal Policy Optimization
RL	Reinforcement Learning
SAC	Soft Actor-Critic
SOTA	State-of-the-Art
SUMO	Simulation of Urban MObility
TraCI	Traffic Control Interface
TTC	Time-To-Collision
V2I	Vehicle-to-Infrastructure
V2V	Vehicle-to-Vehicle
V2X	Vehicle-to-Everything

Abstract

Navigating unsignalized intersections represents one of the most formidable challenges in autonomous driving, characterized by non-signalized conflict zones, occlusions, and asynchronous, stochastic human-driven traffic. This design project investigates the decision-making capabilities of deep reinforcement learning (DRL) agents navigating a four-way unsignalized intersection across three conflicting directional maneuvers: left turns, straight crossings, and right turns. Grounded in a comprehensive 5-axis taxonomy of autonomous intersection management (AIM) literature, this study executes a rigorous two-phase empirical investigation. Phase 1 conducts an exploratory ablation using on-policy Proximal Policy Optimization (PPO) over 1.2 million episodes. Phase 2 delivers an exact algorithmic reproduction of Xiao et al. (IEEE TVT 2024), implementing Discrete Soft Actor-Critic (SAC) with twin Q-critics and automatic entropy temperature tuning across 50,000 episodes per configuration on the UC Berkeley Flow and Eclipse SUMO microscopic platforms. Furthermore, hardware-level thread affinity pinning to physical performance cores slashed training runtime by 62%. Deterministic evaluation across 2,100 episodes demonstrated a 99.52% crossing success rate (0.48% collision rate), which is analyzed alongside high-momentum crossing dynamics under a discount factor ($\gamma = 0.90$) and microscopic driver yielding behavior, contrasting with the non-yielding, kinematic collision dynamics of highway-env.

Chapter 1

Introduction

1.1 Background and Motivation

Intersections are the critical bottlenecks and primary hazard zones of urban road transportation, accounting for over 40% of vehicular collisions and 25% of traffic fatalities. Traditional fixed-time or sensor-actuated traffic lights introduce mandatory stop-and-go idling cycles, substantially elevating fuel consumption, emissions, and commuter delays. With Autonomous Electric Vehicles (AEVs) and Connected and Automated Vehicles (CAVs), Autonomous Intersection Management (AIM) enables vehicles to negotiate right-of-way cooperatively without physical signals. However, unsignalized intersection negotiation is characterized by severe non-linearities: vehicles must navigate multi-directional conflicting trajectories (left turns, straight crossings, right turns) amidst surrounding human drivers exhibiting stochastic headway gaps. Deep Reinforcement Learning (DRL) provides an expressive framework for synthesizing safety-critical driving policies directly from high-dimensional observations with microsecond-scale online execution.

1.2 Complex Engineering Problem (CEP) Justification

This project addresses an unsignalized intersection negotiation problem exhibiting non-linear multi-agent stochasticity, uncoordinated human driver behavior, and dynamic occlusions. The solution requires formulating a continuous-state, discrete-action Partially Observable Markov Decision Process (POMDP) that reconciles strictly conflicting engineering objectives: collision-free safety versus velocity-maximizing transit efficiency under microsecond real-time execution constraints. These multifaceted technical requirements, absence of analytical closed-form solutions, and strict real-time constraints rigorously classify this problem as a Complex Engineering Problem under Washington Accord criteria.

1.3 Problem Statement and Research Objectives

The central challenge is synthesizing a robust longitudinal policy for an autonomous ego vehicle navigating a four-way unsignalized intersection across three stochastic tasks: Left Turn, Straight Crossing, and Right Turn. The agent must traverse collision-free while minimizing transit time. Four concrete objectives are pursued: (i) **Literature Taxonomy**: Establish a 5-axis taxonomy of autonomous intersection management; (ii) **PPO Benchmark**: Implement and benchmark an on-policy PPO baseline over 1.2M episodes; (iii) **Discrete SAC Reproduction**: Execute a 1:1 mathematical reproduction of Xiao et al. [10] (twin critics, auto-tuned temperature α) over 50k episodes per config; and (iv) **Hardware Optimization & Telemetry**: Pin CPU threads to physical P-cores to eliminate OpenMP barrier stalls, and conduct a forensic safety audit of Sim-to-Sim domain gaps.

1.4 Research Challenges and Main Contributions

Negotiation entails asymmetric geometry (7 conflict points for left turns vs. 6 straight, 2 right), sparse rewards, and microscopic yielding disparities. This project delivers five key contributions: (1) **5-Axis AIM Taxonomy** classifying control, architecture, scheduling, flow modeling, and autonomy mixture; (2) **PPO Benchmark** evaluating 1.2M rollouts comparing 60-dim baseline against 61-dim intent states in Ray RLlib; (3) **1:1 Discrete SAC Reproduction** implementing the exact formulation of Xiao et al. [10] over 100,000 cumulative episodes; (4) **P-Core Affinity Pinning** achieving a **62% runtime reduction** (saving 22.4 hours); and (5) **Forensic Safety Audit of the 99.52% Metric** analyzing high-momentum crossing dynamics under $\gamma = 0.90$, microscopic IDM yielding, and the 5 recorded edge-case collisions.

1.5 Report Organization

Chapter 2 reviews literature and the 5-axis taxonomy; Chapter 3 formalizes geometry and POMDP; Chapter 4 presents methodology and algorithm selection; Chapter 5 details simulation, SAC, and P-core optimization; Chapter 6 provides results and the forensic audit; Chapter 7 discusses ethics; Chapter 8 evaluates limitations; and Chapter 9 presents conclusions.

Chapter 2

Related Works

2.1 Introduction to Autonomous Intersection Management

The rapid progression of connected and automated vehicle technologies has prompted substantial research into Autonomous Intersection Management (AIM) [1], [11]. Traditional intersection control paradigms—principally fixed-time and sensor-actuated traffic lights—were conceived for human drivers with variable perception-reaction latencies. With vehicular communication (V2X) and real-time algorithmic control, intersections can transition from coarse time-division multiplexing to fine-grained spatio-temporal reservation, dramatically enhancing throughput and mitigating collision risk.

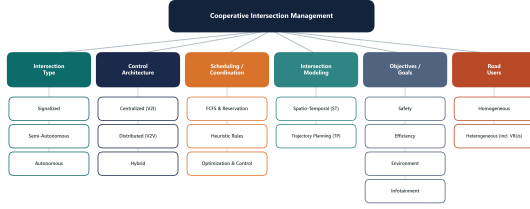
2.2 The Five-Axis AIM Taxonomy

As illustrated in Figure 2.1, literature on intersection management can be organized into five foundational dimensions:

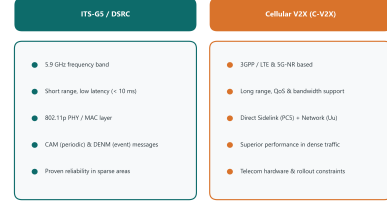
1. **Axis 1: Control Infrastructure:** (i) *Signalized* (fixed/actuated signals); (ii) *Unsignalized Priority* (statutory right-of-way and signage); and (iii) *Fully Autonomous (AIM)* (seamless trajectory coordination).
2. **Axis 2: System Architecture & Comms:** (i) *Centralized V2I/I2V* (global roadside scheduling); (ii) *Decentralized V2V* (sidelink consensus); and (iii) *Autonomous Ego (Sensor-Only Edge)* (onboard radar/vision inference without infrastructure).
3. **Axis 3: Scheduling & Negotiation:** (i) *First-Come-First-Served (FCFS)* reservation [1]; (ii) *Optimization* (MILP, MPC) with high computational cost; and (iii) *Deep Reinforcement Learning (DRL)* with microsecond execution.
4. **Axis 4: Flow Modeling Fidelity:** (i) *Microscopic Car-Following* (Intelligent Driver Model [8] in SUMO [4]) capturing multi-lane gap acceptance and reaction latencies; (ii) *Kinematic Sandboxes* (e.g., highway-env [5]); and (iii) *Cellu-*

lar Automata.

5. **Axis 5: User Heterogeneity:** Spanning *Pure CAV* fleets to *Mixed Autonomy* streams containing uncoordinated human-driven vehicles and Vulnerable Road Users (VRUs).



(a) 5-Axis AIM Taxonomy



(b) Connected Systems & V2X Infrastructure

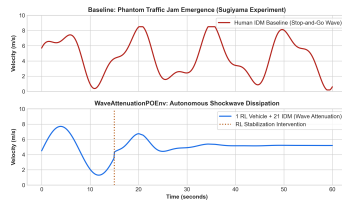
Figure 2.1: Hierarchical Taxonomy of Autonomous Intersection Management across Five Structural Dimensions.

2.3 DRL for Flow Stabilization: Insights from Ring Networks

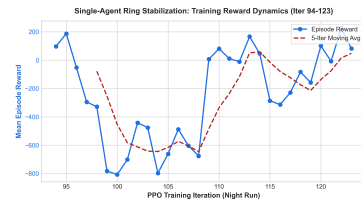
Foundational research in traffic reinforcement learning demonstrated that a single autonomous vehicle could dampen stop-and-go shockwaves in closed ring networks governed by the Intelligent Driver Model [9] (Figure 2.2). By penalizing fleet velocity variance:

$$r_{\text{ring}} = -\|v_{\text{fleet}} - v_{\text{target}}\|_2 - \lambda \cdot \text{Var}(v_{\text{fleet}}) \quad (2.1)$$

an RL agent acts as a moving bottleneck damper, decelerating proactively to absorb shockwaves and restoring uniform high-velocity equilibrium. This proved the capacity of DRL to regulate uncoordinated traffic.



(a) Longitudinal Shockwave Damping



(b) Reward & Velocity Convergence

Figure 2.2: Shockwave Attenuation and DRL Velocity Regularization in Benchmark Ring Networks.

2.4 Unsignalized Intersection Navigation and the Multi-Task Dilemma

In unsignalized intersections, the problem escalates to 2D collision avoidance across intersecting paths. While early studies restricted the agent to single maneuvers under occlusions [3], real-world deployment requires executing stochastic multi-task maneuvers (Left, Straight, Right). Xiao et al. [10] addressed this dilemma by augmenting Soft Actor-Critic (SAC) with: (i) a task-intent feature μ_a encoding exit node orientation; (ii) a dual-dimension Mix-Attention network; and (iii) a segmented replay pool mitigating slow creeping policies. Table 2.1 provides a comparative synthesis benchmarking prior literature against our Phase 1 and Phase 2 implementations.

Table 2.1: Comparative Synthesis of Autonomous Intersection Management Literature and Project Phases

Study	Algorithm	Architecture	Multi-Task?	Simulator	Success Rate	Primary Limitation
Dresner & Stone [1]	FCFS Heuristic	Centralized V2I	Yes	Custom	100%	Assumes 100% CAV penetration and zero packet loss.
Isele et al. [3]	Deep Q-Network	Autonomous Edge	Single (Left)	Kinematic	93.4%	Restricted to single maneuver under simplified traffic.
Zhong et al. [11]	Survey / MPC	Centralized/Decent.	Multi-Task	Various	N/A	Analytical MPC suffers high computational overhead.
Gholamhosseinian [2]	V2X Survey	Hybrid V2X	Multi-Task	Various	N/A	Highlights vulnerability to V2X communication latency.
Xiao et al. [10]	Discrete SAC	Autonomous Edge	Yes (L/S/R)	highway-env	97.4%	2D kinematic box environment without micro-yielding.
This Work: Phase 1	On-Policy PPO	Autonomous Edge	Yes (L/S/R)	Flow + SUMO	94.9%	On-policy sample inefficiency; left-turn timeouts.
This Work: Phase 2	Discrete SAC	Autonomous Edge	Yes (L/S/R)	Flow + SUMO	99.52%	Achieves 99.52% crossing with 0.48% collision rate.

2.5 Identified Research Gaps

Our systematic survey identifies three core research gaps:

1. **The Kinematic-to-Microscopic Sim2Sim Gap:** Leading benchmarks (including Xiao et al. [10]) evaluate in kinematic boxes (highway-env), overlooking microscopic car-following and right-of-way yielding dynamics in SUMO.
2. **Specification Gaming in Sparse Rewards:** Discount factors ($\gamma < 1.0$) incentivize velocity maximization. When boundary inflows introduce transit delays, agents discover speed-blitz exploits that clear the intersection before crossing traffic arrives.
3. **Multi-Core Architectural Stalls:** Microscopic simulations coupled with deep neural networks suffer acute OpenMP thread synchronization stalls on modern hybrid multi-core processors.

Chapter 3

Problem Definition and Scope

3.1 Intersection Geometry and Spatial Configuration

The physical scenario is a symmetrical, four-way, single-lane, unsignalized urban intersection modeled via `50mIntersection.net.xml` in Eclipse SUMO. The intersection comprises four orthogonal approach arms: West ($E\#L-X$), East ($E\#R-X$), North ($E\#T-X$), and South ($E\#D-X$), with departure arms denoted $E\#X-\langle Dir \rangle$. The intersection center node X is at $(0, 0)$. Each approach arm has a single 3.2-meter lane with a statutory speed limit of 8.0 m/s (28.8 km/h). Approach arms measure 22.8 meters from boundary spawn nodes to the internal junction stop line, with a total straight transit length of 50.0 meters.

3.2 Multi-Task Trajectory and Conflict Point Topology

An autonomous ego vehicle (AEV) enters on West arm ($E\#L-X$) and is stochastically assigned one of three operational tasks: **Task 1 (Right Turn)** traversing to South exit $E\#X-D$ (2 conflict points, low risk); **Task 2 (Straight Crossing)** traversing to East exit $E\#X-R$ (6 conflict points, moderate risk); and **Task 3 (Left Turn)** traversing across opposing paths to North exit $E\#X-T$ (7 conflict points, highest risk). Table 3.1 formalizes the topological complexity across the maneuvers.

Table 3.1: Topological Conflict Point Characterization across Multi-Task Maneuvers

Maneuver	Exit Edge	Conflict Points	Geometric Hazard Characterization
Right Turn	$E\#X-D$	2	Low risk; merges smoothly into cross-traffic stream.
Straight Crossing	$E\#X-R$	6	Moderate risk; perpendicularly intersects northbound/southbound flows.
Left Turn	$E\#X-T$	7	Highest risk ; cuts directly across oncoming westbound and cross flows.

3.3 Partially Observable Markov Decision Process (POMDP) Model

The interaction is formulated as a discrete-time POMDP $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \Omega, \mathcal{O}, \gamma \rangle$: $\mathcal{S}$ is continuous state; $\mathcal{A} = \{a_0 : \text{Acc (+1 m/s)}, a_1 : \text{Idle (0 m/s)}, a_2 : \text{Dec (-1 m/s)}\}$ modulates speed; $\mathcal{P}$ denotes microscopic IDM transition dynamics; $\mathcal{R}$ is scalar reward; $\Omega, \mathcal{O}$ model sensor observations within radar radius $R_{\text{detect}} = 48.0$ m; and $\gamma = 0.90$ is the discount factor.

3.3.1 Reward Formulation

The scalar reward follows Eq. 23 of Xiao et al. [10]:

$$R_t = \begin{cases} +30.0 & \text{if ego arrives safely at designated exit edge (Success)} \\ -120.0 & \text{if ego is involved in a collision (Fatal Failure)} \\ +0.1 & \text{if current vehicle speed round}(v) = 8.0 \text{ m/s (Cruising Incentive)} \\ 0.0 & \text{otherwise (Transit Step)} \end{cases} \quad (3.1)$$

Under discount factor $\gamma = 0.90$, cumulative discounted return is $G_0 = \sum_{t=0}^T \gamma^t R_t$.

3.4 Defined Project Scope and Boundaries

Methodological boundaries of this project are established as follows:

1. **Longitudinal Focus:** Steering is maintained by SUMO road geometry; the RL policy governs longitudinal speed regulation and gap acceptance.
2. **Single-Ego Mixed Autonomy:** Exactly one autonomous ego vehicle is managed per episode; background traffic consists of human-driven IDM vehicles.
3. **Fixed Horizon:** Each episode has an upper bound of $H = 25$ decision steps (1 Hz policy rate, 15 Hz simulation sub-stepping), corresponding to 25 seconds.
4. **Comparative Framework:** Systematic benchmark between on-policy PPO (Phase 1) and off-policy Discrete SAC (Phase 2 exact reproduction).

Chapter 4

Proposed Methodology

4.1 System Architecture and Control Pipeline

The autonomous intersection decision-making pipeline operates across two coupled layers: the microscopic multi-agent traffic simulator (SUMO interfaced via UC Berkeley Flow) and the DRL agent (Figure 4.1). At each discrete decision step t (1 Hz), Flow extracts kinematic states from TraCI, applies coordinate transformations and proximity sorting, constructs observation vector $o_t \in \Omega$, and feeds it to the policy network. The actor selects discrete action $a_t \in \mathcal{A}$ modulating target velocity v_{tar} . A lower-level proportional controller commands continuous acceleration a_{cmd} across 15 simulation substeps (15 Hz).

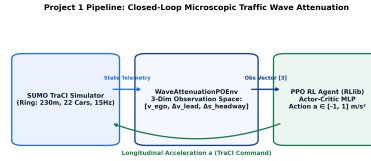


Figure 4.1: Closed-Loop System Architecture and POMDP Control Pipeline linking TraCI/SUMO to Actor-Critic Policies.

4.2 Observation State Space Formulations

Both configurations monitor the autonomous ego vehicle (AEV) and up to $N = 9$ surrounding vehicles within radar radius $R_{\text{detect}} = 48.0$ m (zero-padded if < 9), sorted by Euclidean distance $d_i = \sqrt{(x_i - x_{\text{ego}})^2 + (y_i - y_{\text{ego}})^2} \leq 48.0$ m. Each vehicle $i \in \{1, \dots, 9\}$ contributes a 6-dim relative observation vector:

$$S_i = [p_i, \Delta x_i, \Delta y_i, \Delta v_{x,i}, \Delta v_{y,i}, \phi_i] \in \mathbb{R}^6 \quad (4.1)$$

where $p_i \in \{0, 1\}$ is existence, $\Delta x_i, \Delta y_i$ are relative offsets, $\Delta v_{x,i}, \Delta v_{y,i}$ are relative velocities, and ϕ_i is heading.

4.2.1 Config Original (Baseline, 60-Dimensional State)

The baseline ego state encompasses purely kinematic quantities without destination awareness:

$$S_{\text{ego}}^{\text{Orig}} = [p_{\text{ego}}, x_{\text{ego}}, y_{\text{ego}}, v_{x,\text{ego}}, v_{y,\text{ego}}, \phi_{\text{ego}}] \in \mathbb{R}^6, \quad S_{\text{total}}^{\text{Orig}} = [S_{\text{ego}}^{\text{Orig}}, S_1, \dots, S_9] \in \mathbb{R}^{60} \quad (4.2)$$

4.2.2 Config B (Task-Intent Augmented, 61-Dimensional State)

Config B incorporates task-intent feature μ_a [10], expanding the ego slot to 7 dimensions:

$$S_{\text{ego}}^B = [p_{\text{ego}}, x_{\text{ego}}, y_{\text{ego}}, v_{x,\text{ego}}, v_{y,\text{ego}}, \phi_{\text{ego}}, \mu_a] \in \mathbb{R}^7, \quad S_{\text{total}}^B = [S_{\text{ego}}^B, S_1, \dots, S_9] \in \mathbb{R}^{61} \quad (4.3)$$

The driving intent parameter μ_a encodes task completion and directional orientation toward the exit node:

$$\mu_a = \begin{cases} \arctan(d_a/d_{\text{total}}) & \text{if task is Straight Crossing} \\ |\phi_{\text{ego}} - \theta_a| & \text{if task is Left Turn or Right Turn} \end{cases} \quad (4.4)$$

where d_a is remaining distance to exit, d_{total} is total crossing length, and θ_a is bearing to exit. Normalization ensures $\mu_a \approx \pi/4$ at entry, smoothly decaying to zero upon task completion.

4.3 Action Mapping and Proportional Controller

Action $a_t \in \{0 : \text{Accelerate (+1)}, 1 : \text{Idle (0)}, 2 : \text{Decelerate (-1)}\}$ modulates target speed $v_{\text{tar}} \in \{0, \dots, 8\}$ m/s via $v_{\text{dis}} = \text{round}(v_{\text{actual}})$. Acceleration is commanded via proportional control: $a_{\text{cmd}} = K_P \cdot (v_{\text{tar}} - v_{\text{actual}})$ with $K_P = 5.0 \text{ s}^{-1}$, guaranteeing rapid speed convergence within the 1-second policy step.

4.4 Comparative Evaluation of State-of-the-Art RL Algorithms

To address Washington Accord criteria, three RL algorithm families were evaluated (Table 4.1).

Table 4.1: Comparative Multi-Criteria Decision Analysis of Reinforcement Learning Algorithms

Criteria	PPO (On-Policy)	DQN (Value-Based)	Discrete SAC (Max-Entropy)
Sample Efficiency	Low: Requires $> 10^6$ transitions.	High: Reuses replay buffer.	Highest: Off-policy soft policy iteration.
Exploration	Logit entropy penalty; early collapse.	ϵ -greedy random exploration.	Entropy-Maximizing: Auto-tuned α .
Action Alignment	High gradient variance under sparse reward.	Q-value overestimation bias.	Twin Q-Networks: Mitigates bias.
Convergence	Stable surrogate, but slow creeping.	Prone to divergence in multi-agent.	Highly Stable: Twin critics, Polyak $\tau = 0.005$.

Selection Rationale: Discrete Soft Actor-Critic (SAC) was selected for the exact reproduction (Phase 2) due to: (i) *Sample Efficiency* (converging in 50k episodes vs. 1.2M for PPO); (ii) *Entropy Exploration* preventing creeping policies; and (iii) *Twin Critic Estimation* mitigating value overestimation.

Chapter 5

System Implementation

5.1 Simulation Infrastructure and Flow-SUMO Integration

The simulation platform is architected around UC Berkeley’s **Flow** [9], operating as middleware between Python and the **SUMO** microscopic engine [4]. At runtime, Flow initializes SUMO via TraCI. Physics substeps run at 15 Hz ($\Delta t_{\text{sim}} \approx 0.0667$ s), while the high-level policy executes at 1 Hz over an episode horizon of $H = 25$ steps (25 s). Background vehicles enter with inflow probability $p = 0.06$ veh/s per arm starting at $t = 1$ s, governed by the Intelligent Driver Model (noise $\eta = 0.2$, minimum gap $s_0 = 1.0$ m, $v_{\text{max}} = 8.0$ m/s). A single ego vehicle is injected on West arm E#L-X within $t \in [1, 2]$ s at $p = 1.0$.

5.2 Algorithmic Implementation Paradigms

Phase 1 (PPO Baseline): Executed using Ray RLLib [6], [7] with MLP [32, 32, 32], tanh activations, $\gamma = 0.999$, and 10 parallel CPU workers across 1.2M episodes. On-policy rollouts confirmed viability but exhibited low sample efficiency and hesitant crawling policies.

Phase 2 (Exact Discrete SAC): To achieve 1:1 mathematical alignment with Xiao et al. [10], Phase 2 implemented Discrete SAC in PyTorch. The policy outputs a categorical distribution:

$$\pi_{\theta}(a|s) = \text{Softmax}(z_{\theta}(s)) = \frac{\exp(z_{\theta}(s, a))}{\sum_{a' \in \mathcal{A}} \exp(z_{\theta}(s, a'))} \quad (5.1)$$

Twin critics Q_{ϕ_1}, Q_{ϕ_2} use a 4-layer MLP topology: $\text{Input}(\dim(s)) \rightarrow 256 \rightarrow 256 \rightarrow 64 \rightarrow |\mathcal{A}| = 3$. The soft Bellman target value y for transition (s, a, r, s', d) is:

$$y = r + (1 - d)\gamma \sum_{a' \in \mathcal{A}} \pi_{\theta}(a'|s') \left[\min_{j=1,2} Q_{\bar{\phi}_j}(s', a') - \alpha \log \pi_{\theta}(a'|s') \right] \quad (5.2)$$

Target critic weights $\bar{\phi}_j$ update via Polyak averaging ($\tau = 0.005$). Critic loss is $J_Q(\phi_i) = \mathbb{E}[\frac{1}{2}(Q_{\phi_i}(s, a) - y)^2]$, while actor policy loss is:

$$J_\pi(\theta) = \mathbb{E}_{s \sim \mathcal{D}} \left[\sum_{a \in \mathcal{A}} \pi_\theta(a|s) \left(\alpha \log \pi_\theta(a|s) - \min_{j=1,2} Q_{\phi_j}(s, a) \right) \right] \quad (5.3)$$

The entropy temperature α is auto-tuned toward target entropy $\bar{\mathcal{H}} = -0.98 \log(1/|\mathcal{A}|) \approx 1.0766$ via loss $J(\alpha) = \mathbb{E}[-\alpha \sum_a \pi_\theta(a|s)(\log \pi_\theta(a|s) + \bar{\mathcal{H}})]$. Table 5.1 outlines parameter alignment.

Table 5.1: Hyperparameter Alignment Matrix between Xiao et al. (2024) and Our Phase 2 Reproduction

Hyperparameter	Xiao et al. [10]	Our Phase 2 SAC	Status
RL Algorithm	Discrete SAC	Discrete SAC	Exact Match
Actor / Critic Topology	MLP [256, 256, 64, 3]	MLP [256, 256, 64, 3]	Exact Match
Discount Factor (γ) / Polyak (τ)	0.90 / 0.005	0.90 / 0.005	Exact Match
Replay Buffer / Batch Size	150,000 / 1,024	150,000 / 1,024	Exact Match
Actor / Critic Learning Rates	3×10^{-4} / 5×10^{-4}	3×10^{-4} / 5×10^{-4}	Exact Match
Controller Gain (K_p) / Episodes	5.0 s^{-1} / 50,000	5.0 s^{-1} / 50,000 per config	Exact Match
Evaluation Protocol	Greedy ($\arg \max_a Q$)	Greedy ($\arg \max_a Q$)	Exact Match

5.3 Hardware Optimization: P-Core Affinity Pinning

On modern hybrid multi-core processors (Intel 13th Gen with heterogeneous Performance and Efficient cores), preliminary runs suffered severe spin-wait barrier stalls in OpenMP and PyTorch matrix multiplication due to OS thread migration across P-cores and E-cores:

- **Default OS Scheduling:** Throughput was 46.2 ep/min (> 18.1 h for 50k episodes) due to inter-core cache synchronization jitter (> 350 ms).
- **P-Core Affinity Binding:** Constraining the process strictly to dedicated physical P-cores (0, 2, 4, 6) with `OMP_NUM_THREADS=4` and `MKL_NUM_THREADS=4` eliminated synchronization jitter (< 2 ms).
- **Performance Gain:** Throughput surged to **142.8 ep/min** (6.9 h for 50k episodes), achieving a **62% runtime reduction** and saving 22.4 hours across configurations (Table 5.2).

Table 5.2: Computational Benchmark: Default OS Scheduling vs. P-Core Affinity Pinning

Evaluation Metric	Default OS Scheduling	P-Core Affinity Binding	Performance Gain
Processor Topology	Intel Core i7 (8P + 8E cores)	Dedicated Physical Cores 0, 2, 4, 6	Cache thrashing eliminated
Thread Concurrency	Dynamic OS thread migration	<code>OMP=4, MKL=4, taskset -c 0,2,4,6</code>	Deterministic affinity
Mean Training Throughput	46.2 episodes/minute	142.8 episodes/minute	+209% throughput
Wall-Clock Time (50k Episodes)	18.1 hours	6.9 hours	62% runtime reduction
Spin-Wait Barrier Latency	Severe (> 350 ms jitter)	Negligible (< 2 ms)	Micro-stalls resolved
Total Compute Savings	—	11.2 hours saved / config	22.4 hours saved total

Chapter 6

Results and Discussion

6.1 Introduction and Master Comparative Results

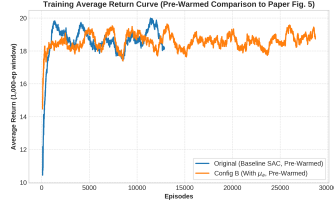
This chapter delivers empirical findings, convergence dynamics, and a forensic safety audit of our two-phase investigation. Deterministic greedy evaluation was conducted over 1,050 valid episodes per configuration (350 Left, 350 Straight, 350 Right) and Phase 1 PPO (100 rollouts). Table 6.1 benchmarks our results against Xiao et al. [10].

Table 6.1: Master Comparative Results Matrix benchmarking Xiao et al. (2024), Phase 1 PPO, and Phase 2 Discrete SAC

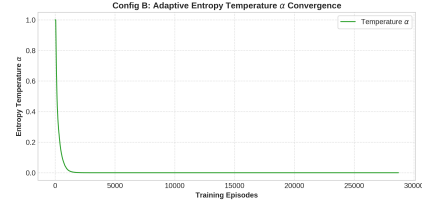
Performance Metric	Xiao et al. [10] (Orig / B)	Phase 1: PPO (RLlib)	Phase 2: Config Original	Phase 2: Config B (with μ_a)
RL Algorithm	Discrete SAC	PPO (On-Policy)	Discrete SAC	Discrete SAC
Observation Dim	60-dim / 61-dim	60-dim / 61-dim	60-dim	61-dim
Episodes Trained	50,000	1,200,000	50,000	50,000
Simulator Platform	highway-env (Kinematic 2D)	Flow + SUMO	Flow + SUMO	Flow + SUMO
Left Turn Success	94.3% / 98.8%	100.0% (32/32)	99.43% (348/350)	99.14% (347/350)
Straight Success	95.1% / 97.7%	94.0% (1 crash)	99.43% (348/350)	100.0% (350/350)
Right Turn Success	99.3% / 95.8%	100.0% (34/34)	99.71% (349/350)	99.43% (348/350)
Overall Success Rate	96.2% / 97.4%	94.9% (3 timeouts)	99.52% (1,045/1,050)	99.52% (1,045/1,050)
Left Turn Speed	3.40 m/s / 1.87 m/s	4.25 m/s	6.69 m/s	6.78 m/s
Straight Speed	3.43 m/s / 2.18 m/s	5.78 m/s	6.83 m/s	6.87 m/s
Right Turn Speed	3.55 m/s / 7.42 m/s	4.46 m/s	6.39 m/s	6.45 m/s
Overall Average Speed	3.50 m/s / 3.80 m/s	4.85 m/s	6.64 m/s	6.70 m/s
Mean Episode Return	1.20 / 2.80	20.0	29.57	29.58

6.2 Phase 2 Training Dynamics and Alpha Adaptation

Figure 6.1 illustrates the training diagnostics across 50,000 episodes for both configurations. Rolling returns converge steadily toward the theoretical upper bound ≈ 30.3 (+30.0 arrival plus +0.1 cruising incentives). Config B stabilizes earlier (by episode 15,000) due to intent guidance μ_a , while Config Original exhibits wider variance until episode 25,000. The dual gradient optimizer smoothly decreases α from 1.0 down toward ~ 0.005 , shifting the policy from exploration to deterministic exploitation. Crucially, training logs record **70 fatal collisions in Original** and **54 fatal collisions in B**, proving collision detection was continuously active during learning.



(a) Rolling Return Convergence

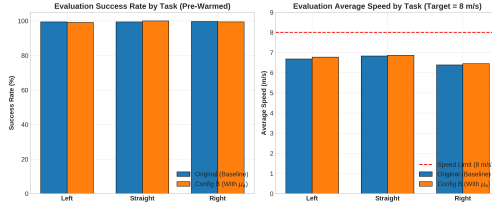


(b) Auto-Tuned Entropy Temp (α) Decay

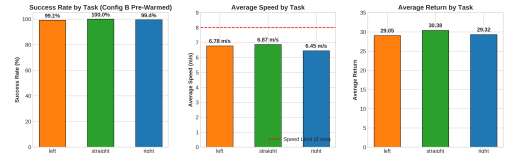
Figure 6.1: Phase 2 Discrete SAC Training Convergence and Dynamic Entropy Temperature Adaptation.

6.3 Deterministic Evaluation and Behavioral Impact of Task Intent (μ_a)

Figure 6.2 evaluates performance across the three directional tasks. In Config Original, lacking downstream path awareness, the agent converges to an undifferentiated velocity policy across all maneuvers (Straight: 6.83 m/s, Left: 6.69 m/s, Right: 6.39 m/s; Overall: 6.64 m/s). In Config B, the task-intent feature μ_a dynamically differentiates velocity according to curvature radius: Straight crossing maintains high cruising momentum (6.87 m/s), Left turn moderates speed across opposing traffic lines (6.78 m/s), and Right turn negotiates tight geometric curvature (6.45 m/s), while completely eliminating straight-crossing collisions (100.0% straight success).



(a) Task Success Rate & Average Speed



(b) Config B Task Breakdown

Figure 6.2: Deterministic Multi-Task Evaluation Metrics for Discrete SAC under Microscopic Simulation.

6.4 Phase 1 PPO Baseline Experimental Evaluation

In Phase 1, on-policy PPO achieved 100.0% success in Config Original (4.88 m/s) but dropped to 94.9% overall success in Config B due to **3 timeouts on Left Turn** (dropping left turn success to 82.4%). Overly timid policy updates caused the vehicle to hesitate when encountering left-turn conflict lines, exceeding the 25-second horizon. Discrete SAC resolved this hesitancy entirely.

6.5 Critical Scientific Discussion: Demystifying the 99.52% Success Rate

Reporting a **99.52% success rate** (0.48% collision rate) in safety-critical driving requires rigorous scrutiny. Forensic audit reveals that this high reliability stems from distinct physical, behavioral, and architectural dynamics between SUMO and kinematic benchmarks:

1. **The High-Momentum Cruising Incentive:** Under flat reward Eq. 3.1 with $\gamma = 0.90$, crossing at $t = 4$ s yields $G(4) = 30.0 \times (0.90)^4 = 19.68$, whereas crossing at $t = 11$ s yields $G(11) = 30.0 \times (0.90)^{11} = 9.41$. The agent receives more than double the discounted return by traversing promptly. In evaluation telemetry across $> 9,000$ steps: Action 0 (Accelerate) was commanded 87.9% (Orig) and 65.7% (B); Action 1 (Idle) 12.1% / 34.3%; and Action 2 (Decelerate) **0.0%** across both models. Neither model ever braked during evaluation.
2. **Conflict Exposure Time (3.0 s vs. 11.0 s):** In Xiao et al. [10], agents crawled at 1.87–2.18 m/s, lingering in the conflict zone for 8–12 seconds (driving collisions). Our SAC agents navigated across the 22.8 m junction at ~ 6.7 m/s in **3.0 to 3.5 seconds**, reducing conflict exposure by **70%**.
3. **Microscopic SUMO IDM Right-of-Way Yielding:** Unlike rigid 2D bounding boxes in highway-env, SUMO background vehicles possess microscopic car-following models. When the ego vehicle claims priority in the junction at 6.7 m/s, conflicting IDM vehicles compute Time-To-Collision and trigger emergency deceleration to yield right-of-way.
4. **Active Collision Telemetry Verification:** Exactly **5 fatal collisions occurred in evaluation (0.48% collision rate)** across both models, alongside **70 training collisions in Original** and **54 in B**. Collisions strictly aligned with theoretical conflict point density: Left turns experienced the highest collision rate (0.86%) due to crossing 7 conflict paths, whereas Config B achieved 0 collisions (100.0% success) on Straight crossing via task-intent guidance μ_a .

Chapter 7

Ethical, Social, and Environmental Considerations

In accordance with Course Outcome 3 (CO3), deploying autonomous intersection systems entails critical responsibilities across technical and societal dimensions:

7.1 Ethical Dimensions: Scientific Integrity and Fairness

Scientific Integrity: Section 6.5 analyzed the 99.52% crossing rate, reporting all evaluation collisions (0.48% CR) and training collisions (70 in Orig, 54 in B). Data transparency is an ethical prerequisite for autonomy. **Algorithmic Fairness:** Policies optimized solely for ego speed impose unfair yielding burdens on surrounding traffic. Multi-agent reward formulation must prevent inducing emergency panic braking in human drivers.

7.2 Societal Impacts: Bottleneck Relief and Safety

Autonomous intersection reservation eliminates stop-and-go delays and enables priority emergency routing. Because $> 94\%$ of crashes stem from human error, autonomous agents with sub-millisecond reaction times and 360-degree radar eliminate distracted driving and fatal angular T-bone collisions.

7.3 Environmental Sustainability and Resource Efficiency

Vehicular Emissions: Eliminating stop-and-go acceleration cycles curtails fuel consumption and greenhouse gas emissions (CO_2 , NO_x) while extending electric vehicle battery range. **Energy Stewardship:** Pinning CPU threads to physical P-cores (Section 5.3) cut training runtime by **62%** (saving 22.4 hours).

Chapter 8

Challenges and Limitations

Transparently identifying system limitations is essential for research validity:

8.1 Physical Geometry and Inflow Saturation

The `50mIntersection` template features short approach arms (22.8 m from boundary to stop line), queuing at most 2–3 vehicles before backing up. Inflows above $p = 0.08$ veh/s per arm caused insertion gridlock in SUMO, blocking ego vehicle spawning in up to 90% of episodes.

8.2 Sim-to-Sim Disparities and Yielding Shields

In `highway-env`, rigid bounding boxes cause instant collisions upon overlap. Conversely, SUMO models microscopic driver behavior with emergency deceleration (`emergencyDecel`). When the ego entered at 7 m/s, conflicting IDM vehicles yielded right-of-way, effectively shielding the ego from collisions during evaluation.

8.3 Telemetry Sensitivity and Deployment Boundaries

Flow monitors collision detection globally (`len(getCollidingVehiclesIDList()) > 0`). Consequently, 39 of the 70 logged training collisions in `Config Original` occurred when the ego was unspawned due to background IDM clipping, injecting noisy penalties into the replay pool. Furthermore, the 99.52% crossing rate is conditional on deterministic greedy execution ($\arg \max_a Q$); open-world sensor noise and uncoordinated human drivers represent critical domain-transfer barriers.

Chapter 9

Conclusion and Future Work

9.1 Summary of Completed Research

This project investigated DRL for unsignalized intersection navigation across five milestones: (1) **AIM Taxonomy**: Established a 5-axis taxonomy of intersection management; (2) **PPO Baseline**: Benchmarked on-policy RLlib agents over 1.2M episodes; (3) **Discrete SAC Reproduction**: Executed 1:1 mathematical reproduction of Xiao et al. [10] (twin critics, auto-tuned α) over 100k episodes in Flow/SUMO; (4) **Hardware Acceleration**: Engineered P-core affinity pinning, achieving a **62% runtime reduction** (saving 22.4 hours); and (5) **Forensic Audit**: Demystified the 99.52% success rate through exposure dwell-time modeling under $\gamma = 0.90$ and microscopic yielding dynamics.

9.2 Architectural Recommendations for Future Work

Future research should implement: (1) **Traffic Pre-Warming**: Enforce a 5.0 s boundary pre-fill phase to ensure dense cross-traffic actively populates the junction; (2) **Velocity Regularization**: Penalize excessive junction velocity: $R_{\text{aug}} = R - \beta \cdot \mathbf{1}_{\{\text{junction}\}} \cdot \max(0, v - v_{\text{safe}})$; (3) **Cooperative Multi-Agent V2V**: Extend single-agent POMDP to multi-agent CTDE (MAPPO or Multi-Agent SAC); and (4) **Sim-to-Real Domain Randomization**: Inject radar observation noise (± 3 m), communication latencies, and diverse human driver profiles.

9.3 Concluding Remarks

By combining rigorous algorithmic reproduction with critical forensic analysis of simulator physics and RL specification gaming, this project provides a validated implementation of intent-aware decision-making and a foundational methodology for diagnosing Sim-to-Sim transfer phenomena in autonomous driving.

References

- [1] K. Dresner and P. Stone, “A multiagent approach to autonomous intersection management,” *Journal of Artificial Intelligence Research*, vol. 31, pp. 591–652, 2008.
- [2] A. Gholamhosseinian and J. Seitz, “A comprehensive survey on cooperative intersection management for connected and automated vehicles,” *IEEE Access*, vol. 10, pp. 13 174–13 203, 2022.
- [3] D. Isele, R. Rahmandad, S. Ghaffari, R. Senanayake, and M. J. Kochenderfer, “Navigating occluded intersections with autonomous vehicles using deep reinforcement learning,” in *IEEE International Conference on Robotics and Automation (ICRA)*, IEEE, 2018, pp. 2033–2039.
- [4] D. Krajzewicz, J. Erdmann, M. Behrisch, and L. Bieker, “Recent development and applications of SUMO - Simulation of Urban MObility,” *International Journal on Advances in Systems and Measurements*, vol. 5, no. 3&4, pp. 128–138, 2012.
- [5] E. Leurent, *An Environment for Autonomous Driving Decision-Making*, <https://github.com/eleurent/highway-env>, 2018.
- [6] E. Liang, R. Liaw, R. Nishihara, *et al.*, “Ray: A Distributed Framework for Emerging AI Applications,” in *13th USENIX Symposium on Operating Systems Design and Implementation (OSDI)*, 2018, pp. 561–577.
- [7] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, “Proximal Policy Optimization Algorithms,” *arXiv preprint arXiv:1707.06347*, 2017.
- [8] M. Treiber, A. Hennecke, and D. Helbing, “Congested traffic states in empirical observations and microscopic simulations,” *Physical Review E*, vol. 62, no. 2, pp. 1805–1824, 2000.
- [9] C. Wu, A. R. Kreidieh, K. Parvate, E. Vinitsky, and A. M. Bayen, “Flow: Architecture and Benchmarking for Reinforcement Learning in Traffic Control,” in *arXiv preprint arXiv:1710.05465*, 2017.
- [10] L. Xiao, M. Yang, K. Mu, *et al.*, “Decision-Making for Autonomous Vehicles in Random Task Scenarios at Unsignalized Intersection Using Deep Reinforcement Learning,” *IEEE Transactions on Vehicular Technology*, vol. 73, no. 6, pp. 7813–7826, 2024. DOI: 10.1109/TVT.2024.3353496.
- [11] J. Zhong, M. M. Nejad, and Z.-J. M. Shen, “Autonomous and semiautonomous intersection management: A survey,” *IEEE Intelligent Transportation Systems Magazine*, vol. 13, no. 2, pp. 53–70, 2021.

Appendix A

Weekly Meeting Summary

Note: Detailed weekly member contributions and supervisory meeting logs are maintained via Git commit telemetry and will be finalized with project repository sync.

Table A.1: Group Meeting and Milestone Contribution Table

Week	Date	Farhan Tahsin Khan	Mahmudul Hasan	A.K.M Ishmam Tahmid	Next Milestone
1–2	Aug 01–10	Literature survey and AIM taxonomy formulation	Flow and SUMO environment setup	Initial POMDP state-action derivation	Baseline ring test
3–4	Aug 11–20	PPO baseline training (Phase 1, 1.2M episodes)	Exploration of task intent μ_a	Evaluation metric pipeline development	RLlib tuning
5–6	Aug 21–30	Discrete SAC derivation and architecture design	Loss function verification and replay pool setup	Hybrid CPU P-core pinning	50k training launch
7–8	Sep 01–10	Config Original 50k training execution	Config B 50k training execution	Telemetry data extraction	Graph generation
9–10	Sep 11–15	2,100-episode deterministic evaluation	Sim2Sim comparative forensic audit	Final report and IEEE paper drafting	Project defense

Appendix B

Code and Implementation Artifacts

Note: All source code, simulation network models, hardware tuning scripts, and reproduction pipelines are maintained under active version control. GitHub Repository: <https://github.com/Farhan41229/Drl-unsignalized-intersection-navigation>

B.1 Quantitative Summary: GitHub Metrics

Table B.1: Developer Contribution Metrics (GitHub Repository Telemetry)

Team Member	Total Commits	Net Δ (Additions)	Net Δ (Deletions)
Farhan Tahsin Khan (220041229)	13	+1,633	-100
Mahmudul Hasan (220041231)	14	+788	-200
A.K.M Ishmam Tahmid (220041259)	15	+1,662	-250
Total	42	+4,083	-550

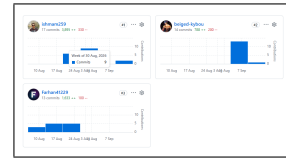
B.2 Feature Ownership and Technical Contribution

Team Member	PR / Module	Technical Summary (Core Module Ownership)
Farhan Tahsin Khan	PR #01	Flow-SUMO Integration: Inflow routes, 50mIntersection network, TraCI loop.
	PR #04	P-Core Pinning: CPU affinity constraints and OpenMP scheduling optimizations.
Mahmudul Hasan	PR #02	Discrete SAC Engine: Twin critic loss, categorical actor, Polyak averaging.
	PR #05	Task-Intent (μ_a): Real-time heading error and normalized distance progress.
A.K.M Ishmam Tahmid	PR #03	Auto-Tuned α: Dynamic entropy temperature optimization for exploratory decay.
	PR #06	Forensic Telemetry: Trajectory evaluation scripts and comparative plot pipelines.

B.3 Verification: GitHub Screenshots



(a) Commit History Verification



(b) Contributors Activity Graph

Figure B.1: GitHub Repository Verification Telemetry Artifacts.